

HASTELLOY C-4 (UNS N06455)

Description:

Hastelloy C-4 is a high-performance nickel-chromium-molybdenum alloy designed for extreme corrosive environments. It offers excellent resistance to oxidizing agents, stress corrosion cracking, and high-temperature oxidation. With low carbon and silicon content, it ensures superior weldability and minimal grain-boundary precipitation, making it ideal for chemical processing, nuclear, pharmaceutical, and environmental industries.

Chemical Composition

Element	Composition (%)
Nickel	≥65.0
Chromium	14.5–16.5
Molybdenum	14–17
Iron	≤5.0
Cobalt	≤2.0
Manganese	≤1
Silicon	≤0.08
Carbon	≤0.015
Phosphorus	≤0.04
Sulfur	≤0.03

Physical Properties

Property	Value
Density	8.64 g/cm ³
Melting Point	1330°C (2430°F)
Tensile Strength	100,000 psi / 690 MPa
Yield Strength	45,000 psi / 310 MPa
Elongation	40%

Mechanical Properties

Property	Metric	Imperial
Tensile Strength (@538°C)	690 MPa	100,000 psi
Yield Strength (@427°C)	310 MPa	45,000 psi
Elastic Modulus (RT)	40%	40%
Hardness, Rockwell B (plate)	89	89

Available Products

- Pipes: Seamless, ERW & EFW
 - Tubes: Capillary, U-Bend
 - Sheets & Plates
 - Coils & Strips
 - Bars: Round, Flat, Hex, Square
 - Pipe Fittings: Elbows, Tees, Reducers, Caps, Stub Ends
 - Flanges: Weld Neck, Slip-On, Blind, Socket Weld, Threaded
 - Fasteners: Bolts, Nuts, Screws, Washers, Studs
 - Custom Fabricated Components
-

Key Advantages

- Superior resistance to strong oxidizers (e.g., nitric acid)
 - Excellent thermal stability up to 1040°C (1900°F)
 - Resistant to intergranular corrosion after welding
 - Maintains mechanical strength in extreme environments
 - High weldability and workability for fabrication
-

Applications

- **Chemical & Petrochemical:** Reactors, heat exchangers, pressure vessels handling hydrochloric, sulfuric, and phosphoric acids
 - **Pharmaceutical & Fine Chemicals:** Equipment exposed to aggressive cleaning agents and high-purity conditions
 - **Pollution Control:** Scrubbers, ducts, stack liners in flue gas desulfurization (FGD) systems
 - **Nuclear Industry:** Heat exchangers, piping in nuclear fuel reprocessing
 - **Pulp & Paper:** Bleaching and chemical recovery sections
 - **Waste Treatment & Environmental Engineering:** Waste acid recovery, incineration scrubbers, and wastewater components
-

This datasheet can now be used like the previous ones for **Hastelloy C276** and **C22**.

I can also **generate a ready-to-download PDF for Hastelloy C-4** if you want, just like we did for the other two alloys. Do you want me to create it?

